

WELLSVILLE, NY, USA

WMO: 725157

Lat: 42.109N

Lon: 77.992W

Elev: 2124

StdP: 13.60

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54757

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.8	3.1	-10.1	3.4	1.3	-5.7	4.4	5.4	26.9	22.2	25.0	21.8	10.4	300	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	84.0	69.3	81.7	68.1	79.4	66.8	72.1	79.9	70.6	78.0	69.2	76.4	8.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.9	119.0	76.3	68.2	112.2	74.6	66.5	105.5	73.2	37.2	79.7	35.7	77.7	34.6	76.6	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	20.4	18.3	DB	-6.6	87.9	5.0	3.3	-10.2	90.3	-13.1	92.2	-15.9	94.1	-19.5	96.5
			WB	-7.3	74.6	4.9	1.5	-10.8	75.7	-13.6	76.6	-16.3	77.5	-19.9	78.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	22.9	24.5	32.8	45.2	56.5	63.6	68.3	66.7	60.4	49.4	38.6	29.0
	DBStd	18.49	12.32	11.10	11.78	10.41	9.11	6.51	5.47	5.42	7.16	9.41	9.74	9.83
	HDD50	3502	841	715	548	212	39	1	0	0	4	130	358	654
	HDD65	7073	1305	1133	998	597	291	101	30	46	176	489	791	1116
	CDD50	2271	2	2	15	69	241	410	566	518	316	112	17	3
	CDD65	366	0	0	0	4	27	61	131	99	38	6	0	0
	CDH74	2280	0	0	0	42	211	387	828	551	239	22	0	0
	CDH80	391	0	0	0	4	28	60	169	88	41	1	0	0
Wind	WSAvg	8.4	10.3	10.3	10.0	9.7	7.8	7.0	6.2	5.9	6.3	8.4	9.2	10.0
Precipitation	PrecAvg	39.0	2.6	1.9	2.9	3.1	3.2	4.0	4.1	4.0	3.9	3.5	3.1	2.7
	PrecMax	48.7	4.2	4.2	4.9	5.1	6.6	6.7	8.3	7.7	6.8	7.0	5.3	4.8
	PrecMin	29.4	0.7	0.8	1.5	1.7	1.1	0.4	1.8	1.9	2.0	1.1	1.1	1.6
	PrecStd	5.2	0.9	0.9	0.9	0.9	1.5	1.7	1.7	1.5	1.4	1.6	1.1	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	61.2	69.4	79.5	83.2	84.9	88.0	85.9	84.9	76.9	67.1	59.8
		MCWB	52.6	53.3	55.5	58.2	66.0	69.7	70.1	71.3	70.0	64.7	55.1	54.5
	2%	DB	51.8	52.0	61.4	73.3	79.8	81.8	84.2	82.4	80.1	72.3	61.2	52.7
		MCWB	47.6	45.9	50.3	55.8	63.9	68.7	69.9	68.7	67.5	61.3	52.3	48.5
	5%	DB	45.8	45.7	55.7	67.9	75.8	78.8	81.4	79.9	76.6	68.3	57.2	47.4
		MCWB	41.4	39.5	48.3	53.1	62.3	66.9	69.3	67.6	65.7	60.4	50.1	43.6
	10%	DB	40.4	40.2	51.0	62.9	71.7	75.3	78.8	77.1	72.7	63.9	54.1	42.8
		MCWB	36.8	35.7	43.5	50.7	59.7	64.7	67.3	66.4	63.8	57.1	47.9	39.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	54.1	57.2	62.1	69.5	72.5	74.3	73.6	73.0	68.4	58.9	54.9
		MCDB	56.6	59.8	67.2	74.0	79.9	80.8	82.9	82.0	80.4	73.5	63.2	59.0
	2%	WB	48.3	46.5	53.3	58.5	66.8	70.2	72.3	71.5	69.6	64.7	54.4	48.9
		MCDB	50.9	50.2	58.9	68.5	75.8	78.8	80.5	78.1	77.7	69.6	58.3	51.9
	5%	WB	42.1	41.0	49.4	55.5	64.3	68.4	70.8	69.9	67.6	61.5	51.2	43.9
		MCDB	45.1	45.1	55.2	64.6	72.0	76.1	78.3	76.9	74.1	66.5	56.3	46.9
	10%	WB	36.7	35.5	44.1	52.6	62.2	66.8	69.2	68.2	65.4	57.5	48.2	39.3
		MCDB	40.3	40.2	50.3	61.2	69.3	73.4	75.9	74.7	71.0	63.2	53.3	42.3
Mean Daily Temperature Range	5% DB	MDBR	13.0	14.2	16.3	19.2	19.2	17.8	17.9	17.7	17.8	15.3	13.6	11.0
		MCDBR	17.1	19.4	22.6	26.1	22.3	20.4	19.8	19.9	20.8	19.8	18.6	16.7
		MCWBR	15.1	15.5	14.3	14.4	11.0	10.9	9.5	9.5	10.3	11.2	13.0	14.3
	5% WB	MCDBR	17.4	19.4	20.1	22.5	19.2	17.3	16.8	16.5	17.3	17.2	16.6	16.4
MCWBR		16.5	17.0	14.8	14.6	10.4	10.2	9.1	8.8	9.5	11.6	13.6	15.4	
Clear-Sky Solar Irradiance	taub	0.289	0.306	0.326	0.354	0.379	0.421	0.421	0.413	0.353	0.339	0.300	0.282	
	taud	2.381	2.332	2.382	2.367	2.325	2.225	2.229	2.242	2.403	2.450	2.525	2.473	
	Ebn at Noon	272	285	292	290	284	271	270	268	279	270	269	267	
	Edh at Noon	25	31	34	37	40	44	43	42	33	28	22	22	
All-Sky Solar Radiation	RadAvg	448	709	1053	1397	1647	1769	1800	1610	1297	818	527	360	
	RadStd	39	82	108	152	168	149	141	97	141	85	73	51	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page